Odor-weighted maps of APL synapses

> **QUESTION**
>
> How does each APL synapse in the calyx get an odor response, what do the odor-weighted maps look like in 2D and 3D, and does weighting by APL's synapses change how similar two odors look?

INPUTS

- Yang's ORN responses `responses_wide.csv` (2026-11-03), turned into fly-mean tables in `orn_inputs_2026-11-03/` : dataset 1 (megamat + diagnostics, 9 flies) and dataset 4 (validation2 + diagnostics, 6 flies)
- Yang's GH146 PN imaging `GH146_ij_certain-roi_stats.csv` (7 flies, 29 glomeruli, 41 odor presentations)
- PN ↔ APL calyx synapses of the corrected calyx (`with_outside_ROI_synapses/` , Report 02)
- KC ↔ APL synapses on the new claws (`with_outside_ROI_synapses/claws_v2m2/` , Report 02); hemibrain and male CNS only

OUTPUTS

- `with_outside_ROI_synapses/odor_weighted/new_ORN_2026-11-03/` : 2D maps `<set>_map_<kind>_posterior_view.png` , 3D maps `<set>_3D_maps_{PN,KC}_APL.pdf` , similarity figures and tables `odor_similarity_{PN,KC}_APL_weighted_vs_once.*` , heat maps `<set>_odor_distance_heatmaps_{PN,KC}_APL.png` with `odor_distance_{PN,KC}_APL_maps.csv` ; `README.md` there

SCRIPTS

- `scripts/orn_inputs_2026_11_03.py` (ORN tables, consistency check)
- `scripts/odor_maps_2026_11/` : `common.py` , `maps2d.py` , `maps3d.py` , `kc_maps.py` (KC and KC + PN maps), `similarity.py` (PN), `kc_similarity.py` (KC)
- `scripts/reports/r03_odor_maps.py` (this report)

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The maps show where in the calyx APL meets the PN boutons and KC claws that a given odor drives. The step that gives each synapse a glomerulus and so an odor response is also how odors enter the model (Report 05), so it is described here in full.

What changed in this version (2026-11-04)

New ORN data. The ORN responses come from Yang's `responses_wide.csv`. Dataset 1 (megamat + diagnostics) and dataset 4 (validation2 + diagnostics) are kept separate, because their diagnostic odors failed the consistency check (Section 1.2).

Corrected calyx everywhere: the synapses outside the official calyx ROI are included (Report 02).

New claws (8 µm along the KC skeleton, split by PN, ≥ 2 PSDs per PN; Report 02). They exist for hemibrain and male CNS, so the KC maps cover the three calyces of the model.

New maps. There is an odor-weighted APL → KC + PN map, and every map now also comes in 3D (two angles per odor and calyx; BANC left out).

New analyses. Odor similarity weighted by synapses is now cosine (Section 5), and odor-by-odor distance heat maps have been added (Section 6).

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range
g	An antennal-lobe glomerulus (FlyWire names; hemibrain VC3l, VC3m, VC5 renamed VC3, VC5, VM6)	40 in the ORN tables, 29 in GH146
o	An odor presentation: odor name and log10 dilution, e.g. "pa @ −3"	
f	A fly	
$r_{g,o}^{(f)}$	Response of glomerulus g to odor o in fly f, averaged over that fly's trials. ORN: as given in `responses_wide.csv`; GH146: peak ΔF/F	ORN: units as in Yang's file (not stated there); GH146: ΔF/F
$\bar{r}_{g,o}$	Fly-mean response (Equation 1)	same units
e	One APL synapse in the calyx	
$g(e)$	Glomerulus of synapse e: of its PN (PN ↔ APL), or of its claw's PN (KC ↔ APL on a claw)	
$w_o(e)$	Weight of synapse e for odor o: $\bar{r}_{g(e),o}$	blank if $g(e)$ has no response
n_g	Number of synapses (of one kind, in one calyx) whose glomerulus is g	count
S_{ab}	Cosine similarity of odors a and b (Equations 4, 7)	−1 to 1
σ	Width of the 3D Gaussian that smooths a synapse map (Section 6)	0, 2 (main) or 5 µm
K_{ab}	Overlap of the smoothed maps of odors a and b (Equation 5)	response² units
d_{ab}	Euclidean distance between the smoothed maps of odors a and b (Equation 6)	response units

1 Odor responses per glomerulus

1.1 Three response tables

Each response table is averaged in two steps. First each fly's trials are averaged, then the flies:

$$\bar{r}_{g,o} = \frac{1}{|\mathcal{F}_{g,o}|} \sum_{f \in \mathcal{F}_{g,o}} r_{g,o}^{(f)} \tag{1}$$

$\mathcal{F}_{g,o}$ = the flies in which glomerulus g was imaged with odor o (missing values are skipped).

Table 2. The three response sets used for the maps. In dataset 1, 2h, ms and va @ −3 were given in both megamat and diagnostics, and the megamat presentation is used. GH146 has ms @ −3 in both panels; the two are kept and marked " (diag)" and "(megamat)". The model uses a 16-odor ORN table built from the same two datasets (`ORN_TNT16_flymean.csv` : 12 megamat odors from dataset 1, and paa @ −4, −aPine, euc and B-myr from dataset 4).

Response set	flies	glomeruli	odor presentations	range of $\bar{r}_{g,o}$	missing
ORN, dataset 1 (megamat + diagnostics)	9	38	17 megamat + 25 diagnostics	−0.18 to 1.63	0.0%
ORN, dataset 4 (validation2 + diagnostics)	6	40	22 validation2 + 25 diagnostics	−0.31 to 2.44	0.1%
GH146 PNs (diagnostics + megamat)	7	29	24 diagnostics + 17 megamat	−0.16 to 1.96	0.0%

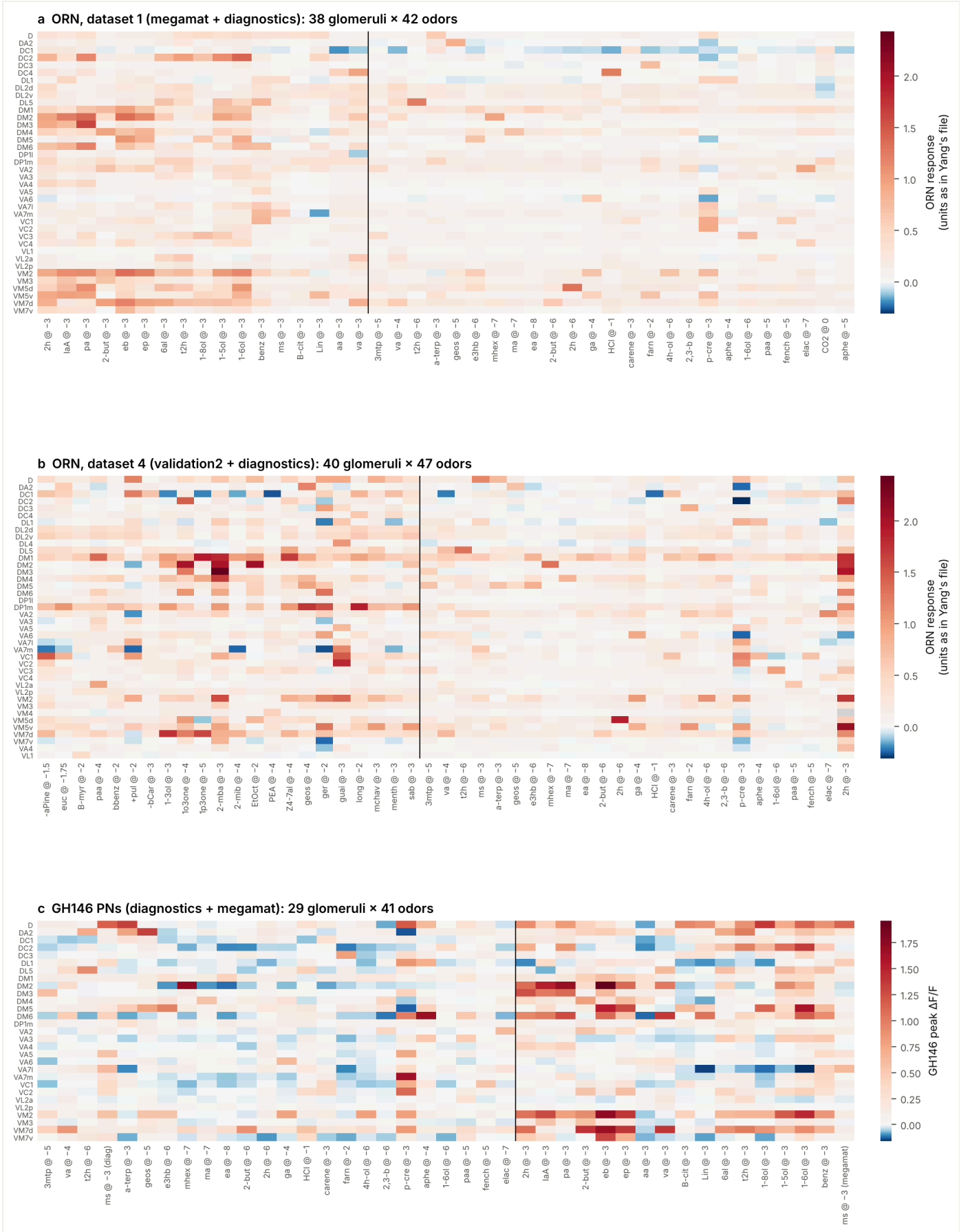


Figure 1. The response tables. Rows are glomeruli and columns are odor presentations. The vertical line separates the first panel (megamat or validation2; diagnostics in GH146) from the second. Color is the fly mean (Equation 1) on a red–white–blue scale whose white is exactly 0. The blue end is the smallest value and the red end the largest: of both ORN tables for panels a and b (one shared scale), and of the GH146 table for panel c. Grey cells were not measured (DL4 and VM4 were imaged only in dataset 4).

1.2 Are the two ORN diagnostic sets the same? No, so they are kept apart

Yang asked to merge the diagnostic odors of the two ORN datasets only if they are consistent. The check compares 3 flies of dataset 1 with 3 flies of dataset 4, and 3 flies with 3 other flies of the same dataset. It does this for the response pattern and for the odor-by-odor similarity matrix. The two count as the same only if (1) the between-dataset correlation is not below the 5th percentile of the within-dataset ones, and (2) a fly-label permutation test finds no difference.

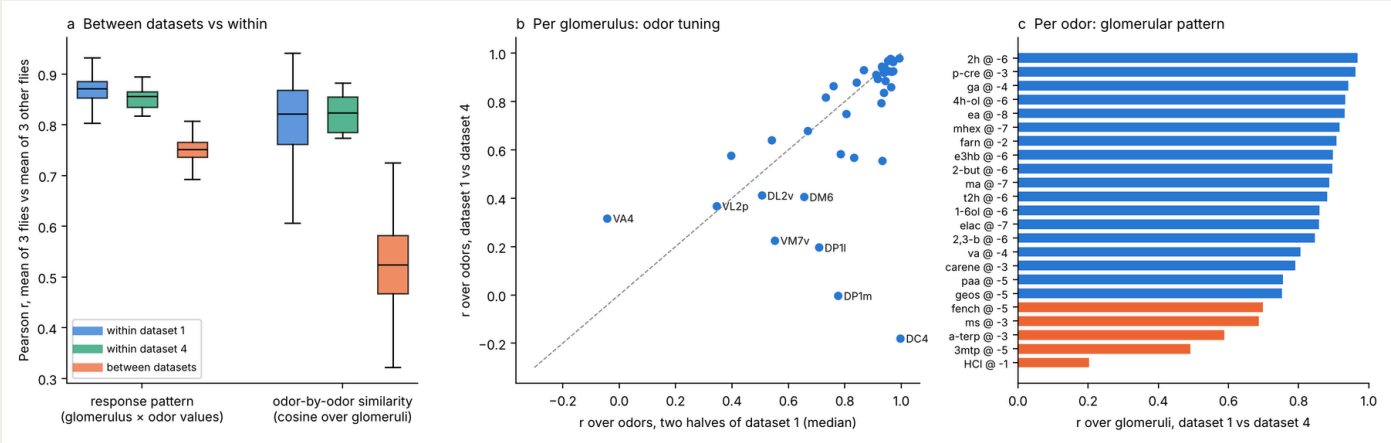


Figure 2. The two diagnostic sets fail the consistency check. Panel a: correlations within dataset 1, within dataset 4 and between the datasets, for the response pattern and for the odor-similarity matrix (3 flies against 3 flies, many random splits). Panels b and c: the glomeruli and odors where the two datasets differ most. Details in `orn_inputs_2026-11-03/README.md`.

- Response pattern: median r 0.76 between datasets, against 0.86–0.87 within. Odor similarity: 0.53 between, against 0.82 within. The fly-label permutation gives p = 0.0002.
- The largest differences: HCl @ −1 drives DC4 strongly in dataset 1 (1.24) but hardly in dataset 4 (0.17); DM5 responds to a-terp and 3mtp only in dataset 4; DC4, DP1m and DP1l are tuned differently. Leaving HCl out does not rescue the check (odor similarity r 0.60).
- So every map and heat map below is made separately for **dataset 1** (megamat + diagnostics) and **dataset 4** (validation2 + diagnostics).

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2 Giving each synapse a glomerulus

- **PN ↔ APL synapses.** The PN's cell type names its glomerulus (e.g. `DM1_lPN` → DM1). Only uniglomerular types count. Types that name two glomeruli (VP1d+VP4, VP1m+VP5, VP3+VP1l) count for both. "X+" types (VP2+, VP3+) count for X. Multiglomerular (M_) and untyped PNs get no glomerulus.
- **KC ↔ APL synapses.** Only synapses **on a claw** get a glomerulus: that of the claw's PN. With the new claws (Report 02) a KC ↔ APL synapse is on a claw when it lies within 8 µm, along the KC, of a PN synapse of that claw. 56–58% of KC → APL and 66–69% of APL → KC calyx synapses are on a claw. A claw whose PN type names two glomeruli has no single glomerulus and stays grey.
- **APL → KC + PN** (Yang's new map) puts APL's two outputs in the calyx together: APL → KC synapses on claws and APL → PN synapses on boutons.

Each synapse e then carries, for each odor, the response of its glomerulus:

$$w_o(e) \;=\; \bar{r}_{g(e),\,o} \tag{2}$$

Blank (drawn grey) when $g(e)$ is not in the response table, or e has no single glomerulus.

Table 3. How many synapses can be colored. Corrected calyx; both directions together for PN ↔ APL. "Colored" = the synapse's glomerulus has a response in that table. The rest belong to glomeruli that were not imaged (mostly thermo- and hygrosensory VP glomeruli), to multiglomerular PNs, or to claws whose PN names two glomeruli. GH146 covers fewer glomeruli (29) than the ORN tables (40).

Calyx	Synapses	number	colored, ORN dataset 1	colored, ORN dataset 4	colored, GH146
FlyWire 783 L	PN ↔ APL	4,369	80%	81%	63%
FlyWire 783 R	PN ↔ APL	5,748	79%	81%	61%
Hemibrain R	PN ↔ APL	5,673	74%	77%	59%
Male CNS L	PN ↔ APL	5,323	77%	79%	63%
Male CNS R	PN ↔ APL	4,401	73%	77%	57%
Hemibrain R	KC → APL on claws	10,795	78%	79%	62%
Hemibrain R	APL → KC on claws	17,965	78%	79%	62%
Hemibrain R	APL → KC + PN	20,868	77%	79%	61%
Male CNS L	KC → APL on claws	11,406	79%	81%	66%
Male CNS L	APL → KC on claws	14,353	80%	82%	64%
Male CNS L	APL → KC + PN	16,925	80%	82%	64%
Male CNS R	KC → APL on claws	9,801	78%	80%	62%
Male CNS R	APL → KC on claws	11,335	77%	79%	60%
Male CNS R	APL → KC + PN	13,271	77%	79%	60%

3 2D maps

Each panel shows one calyx in posterior view (Report 02, Table 4). Every calyx synapse of the chosen kind is a dot at its position, colored by $w_o(e)$; synapses without a response are grey. The color map is `RdBu_r` with a two-slope normalization, so 0 is white, the smallest value $v_{\min}$ is the darkest blue and the largest value $v_{\max}$ the darkest red:

$$c(v) \;=\; \mathrm{RdBu_r}\big(t(v)\big), \qquad t(v) = \begin{cases} \frac{1}{2} + \frac{1}{2}\, v / v_{\max} & v \geq 0 \\ \frac{1}{2} - \frac{1}{2}\, v / v_{\min} & v < 0 \end{cases} \tag{3}$$

One scale for both ORN tables ($v_{\min}$ = −0.31, $v_{\max}$ = 2.44) and one for GH146 (−0.16, 1.96). Within a panel, the strongest colors are drawn on top.

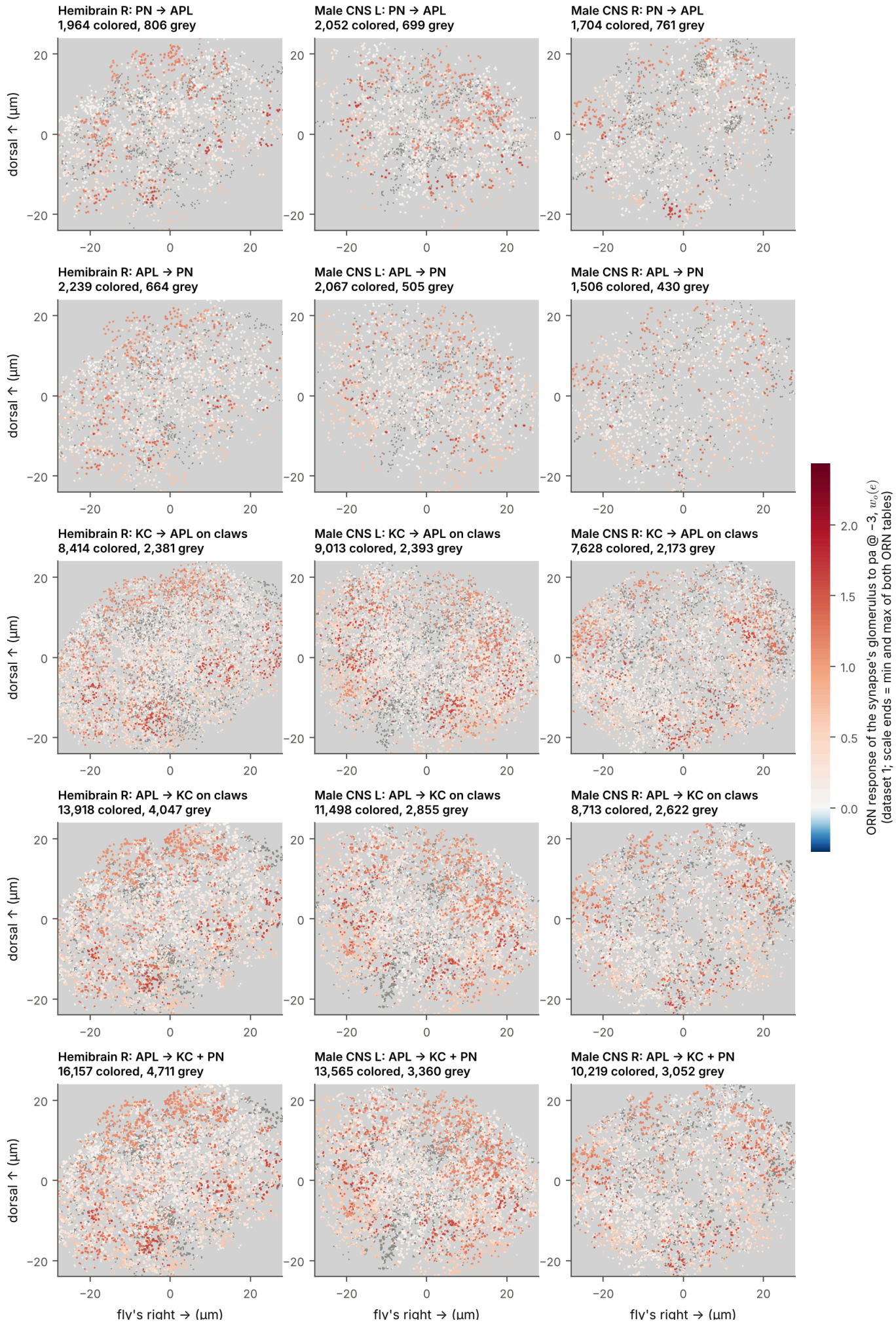

Hemibrain R: PN → APL
1,964 colored, 806 grey
Male CNS L: PN → APL
2,052 colored, 699 grey
Male CNS R: PN → APL
1,704 colored, 761 grey
Hemibrain R: APL → PN
2,239 colored, 664 grey
Male CNS L: APL → PN
2,067 colored, 505 grey
Male CNS R: APL → PN
1,506 colored, 430 grey
Hemibrain R: KC → APL on claws
8,414 colored, 2,381 grey
Male CNS L: KC → APL on claws
9,013 colored, 2,393 grey
Male CNS R: KC → APL on claws
7,628 colored, 2,173 grey
Hemibrain R: APL → KC on claws
13,918 colored, 4,047 grey
Male CNS L: APL → KC on claws
11,498 colored, 2,855 grey
Male CNS R: APL → KC on claws
8,713 colored, 2,622 grey
Hemibrain R: APL → KC + PN
16,157 colored, 4,711 grey
Male CNS L: APL → KC + PN
13,565 colored, 3,360 grey
Male CNS R: APL → KC + PN
10,219 colored, 3,052 grey
dorsal ↑ (µm)
fly's right → (µm)
20
0
−20
2.0
1.5
1.0
0.5
0.0
ORN response of the synapse's glomerulus to pa @ −3, $w_o(e)$
(dataset 1; scale ends = min and max of both ORN tables)

Figure 3. One odor drawn on APL's calyx synapses. Odor = pentyl acetate at 10^{-3} (pa @ −3), ORN dataset 1. Columns are the three calyces of the model. Rows are the kinds of APL synapse: PN → APL and APL → PN (all calyx synapses of uniglomerular PNs); KC → APL and APL → KC (only synapses on a new claw); and APL → KC + PN (rows 2 and 4 together). Each dot is one synapse in posterior view, in µm from the calyx center (the midpoint of the 0.2–99.8% range of PN ↔ APL positions). Color = response of the synapse's glomerulus (Equation 2) on the scale of Equation 3; grey = no response or no single glomerulus. The panel background is light grey so that white (0) stays visible.

Table 4. The full 2D map sets. In `with_outside_ROI_synapses/odor_weighted/new_ORN_2026-11-03/`. `<set>` is `ORN_megamat_diag` (dataset 1, 42 odors), `ORN_valid2_diag` (dataset 4, 47 odors) or `GH146` (41 odors). Each file has one row per odor in Yang's order (first panel, then diagnostics marked "·d").

Kind	Calyces (columns)	File
PN → APL, APL → PN	all five calyces of FlyWire, hemibrain and male CNS	`<set>_map_{PN_to_APL,APL_to_PN}_posterior_view.png`
KC → APL, APL → KC (on claws)	hemibrain R, male CNS L and R	`<set>_map_{KC_to_APL,APL_to_KC}_posterior_view.png`
APL → KC + PN	hemibrain R, male CNS L and R	`<set>_map_APL_to_KC_PN_posterior_view.png`

As before, the strongest boutons and claws of an odor are spread over the whole calyx rather than gathered in one place. The KC maps look like the PN maps. Each claw takes the response of its PN, and claws sit around their PN's boutons, so both kinds of synapse carry the same glomerular pattern at the same places. The APL → KC + PN map is dominated by APL → KC synapses (85%–86% of the dots).

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4 3D maps

The 3D maps place every synapse at its position in anatomical axes: the fly's right, anterior and dorsal, in µm from the calyx center. "Anterior" is signed so that the peduncle lies anterior of the calyx in every dataset. Each page is one odor. Every calyx is shown from two angles: **A**, posterior-lateral (from behind, turned 35° toward the calyx's lateral side, 20° up), and **B**, lateral (from the calyx's lateral side, 20° up). The projection is orthographic, so sizes do not shrink with depth. Grey dots are drawn first and colored ones on top. BANC is left out, as Yang asked.

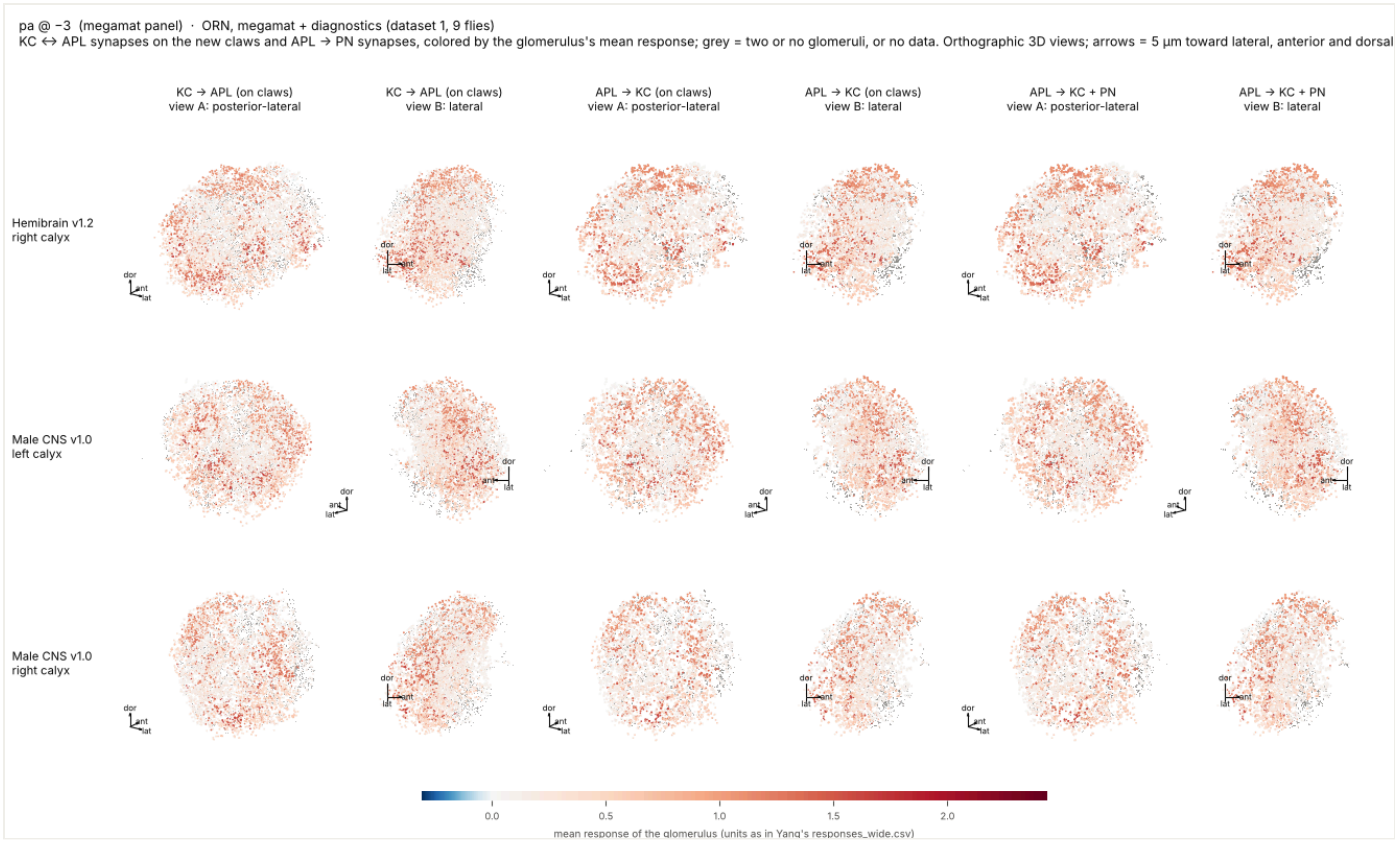


Figure 4. A 3D map of KC ↔ APL synapses on claws, and of APL → KC + PN. pa @ −3, ORN dataset 1. Rows: hemibrain R, male CNS L, male CNS R. Columns: KC → APL on claws, APL → KC on claws, and APL → KC + PN, each from angles A and B. The small arrows are 5 µm long and point lateral, anterior and dorsal. One page of `ORN_megamat_diag_3D_maps_KC_APL.pdf`.



Figure 5. A 3D map of PN ↔ APL synapses. The same odor for PN → APL and APL → PN synapses in the five calyces of FlyWire, hemibrain and male CNS (rows), each from angles A and B. One page of `ORN_megamat_diag_3D_maps_PN_APL.pdf`.

The PDFs (`<set>_3D_maps_PN_APL.pdf` and `<set>_3D_maps_KC_APL.pdf` in `with_outside_ROI_synapses/odor_weighted/new_ORN_2026-11-03/`) have one page per odor for each of the three response sets.

5 Does weighting by APL's synapses change odor similarity?

If APL's synapses favored some glomeruli, two odors could look more or less alike "through APL's synapses" than through the glomeruli alone. Each odor becomes a vector over the synapses of one kind in one calyx, with entry $w_o(e)$; synapses without a response are left out. The similarity of two odors is the cosine of their vectors (as in Yang's RDMs). All synapses of a glomerulus carry the same value, so this equals a cosine over glomeruli in which glomerulus g counts n_g times:

$$S_{ab}^{\text{weighted}} = \frac{\sum_g n_g \, \bar{r}_{g,a} \, \bar{r}_{g,b}}{\sqrt{\sum_g n_g \, \bar{r}_{g,a}^2} \, \sqrt{\sum_g n_g \, \bar{r}_{g,b}^2}} \tag{4}$$

Setting every $n_g = 1$ gives S_{ab}^{once}, glomeruli counted once. ORN: the 16 TNT odors of `ORN_TNT16_flymean.csv` (120 pairs); GH146: the 13 of them that GH146 has (78 pairs). Only glomeruli with a value for every odor are used (DL4 and VM4 lack 12 of the 16 ORN odors and are left out).

Yang asked for six panels, "3 ORN + 3 GH146". These are the three calyces of the model (hemibrain R, male CNS L, male CNS R), for the ORN and the GH146 responses. The analysis is done once for PN ↔ APL synapses and once for KC ↔ APL synapses on claws.

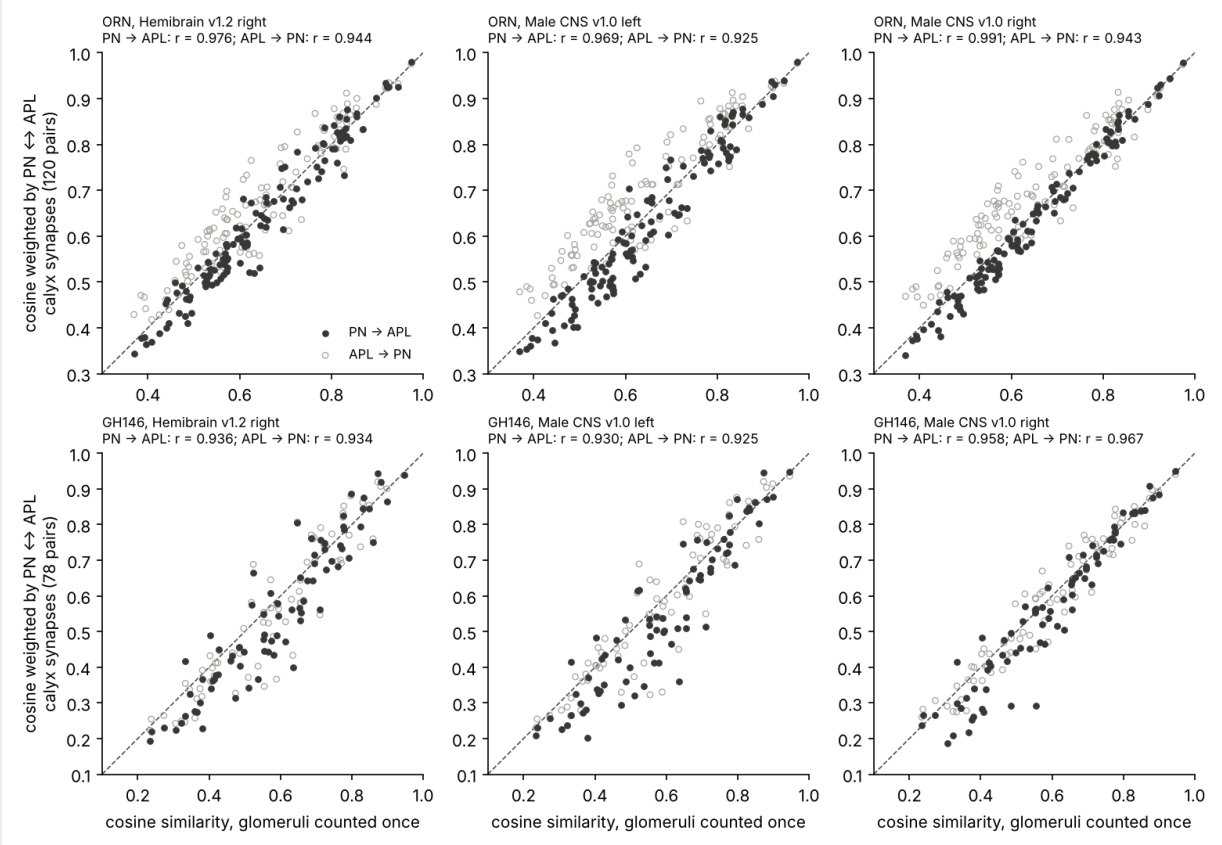


Figure 6. Odor similarity weighted by PN ↔ APL synapses, against glomeruli counted once. Each dot is one pair of odors. x: cosine similarity with every glomerulus counted once. y: the same, each glomerulus weighted by its PN → APL synapses (filled) or APL → PN synapses (open) in that calyx. Dashed line: equal. r in the titles = correlation of x and y over pairs. The two FlyWire and two BANC calyces are in `odor_similarity_PN_APL_weighted_vs_once.csv`.

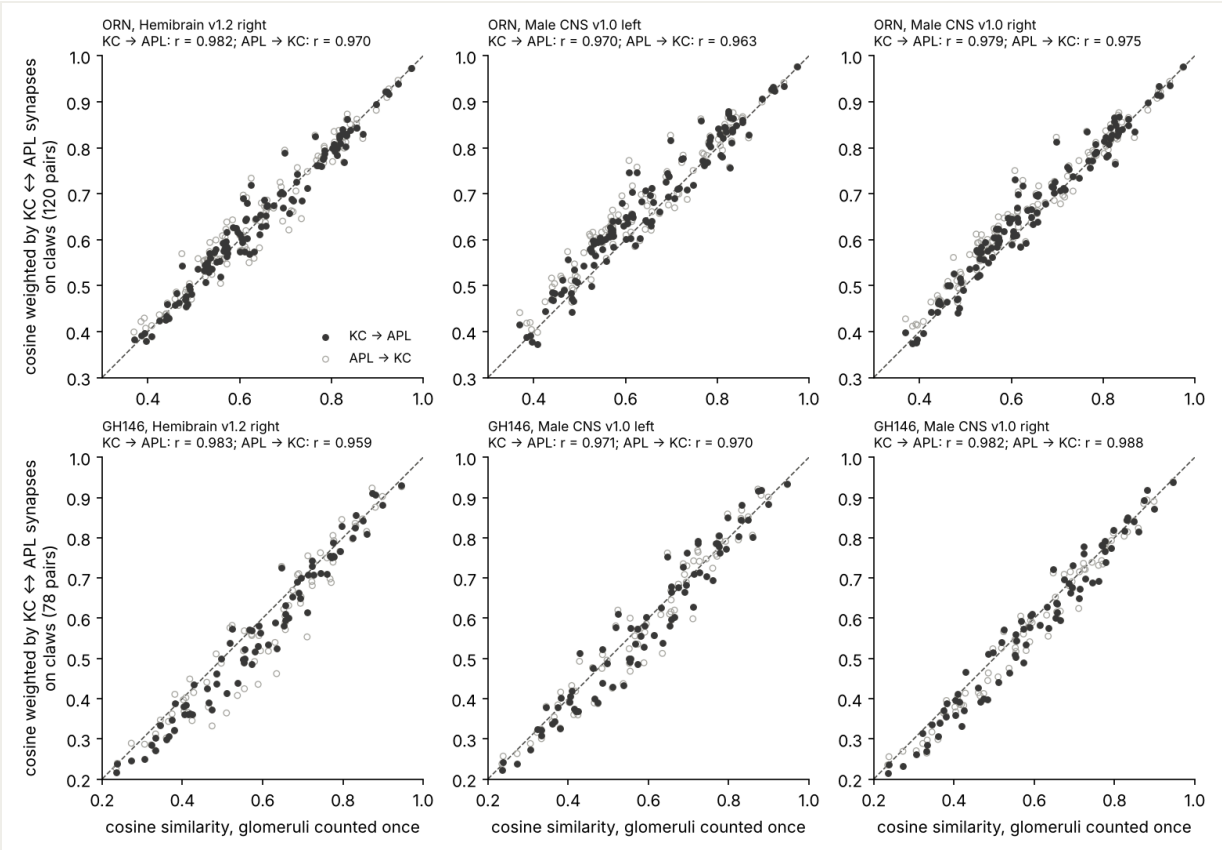


Figure 7. Odor similarity weighted by KC ↔ APL synapses on claws. As Figure 6, with the glomerulus weights n_g = KC → APL synapses (filled) or APL → KC synapses (open) on that glomerulus's claws (new claws). The APL → KC + PN weighting is in `odor_similarity_KC_APL_weighted_vs_once.csv`.

Table 5. How much weighting changes odor similarity (three model calyces). Ranges over hemibrain R, male CNS L and male CNS R. "Mean change" = mean over pairs of $S^{\text{weighted}} - S^{\text{once}}$.

Responses	Weights n_g from	r, weighted vs once	mean change of S_{ab}	largest change of one pair
ORN	PN → APL	0.969–0.991	−0.023 to −0.013	0.11
ORN	APL → PN	0.925–0.944	+0.029 to +0.050	0.17
ORN	KC → APL on claws	0.970–0.982	+0.003 to +0.025	0.14
ORN	APL → KC on claws	0.963–0.975	+0.006 to +0.034	0.16
ORN	APL → KC + PN	0.960–0.973	+0.009 to +0.036	0.17
GH146	PN → APL	0.930–0.959	−0.045 to −0.033	0.27
GH146	APL → PN	0.925–0.967	−0.021 to −0.004	0.26
GH146	KC → APL on claws	0.971–0.983	−0.030 to −0.011	0.11
GH146	APL → KC on claws	0.960–0.988	−0.030 to −0.008	0.17
GH146	APL → KC + PN	0.958–0.987	−0.029 to −0.008	0.18

- **Weighting barely changes odor similarity**, for PN and KC synapses alike: weighted and unweighted similarities correlate at r ≥ 0.93 over odor pairs in every case (≥ 0.96 for KC synapses on claws).
- **The direction of the small shifts depends on the responses.** With ORN responses, weighting by APL's outputs (APL → PN, APL → KC) raises similarities slightly. With GH146 every weighting lowers them slightly. Weighting by PN → APL lowers them with both.

- So the glomerular distribution of APL's synapses does not by itself reshape odor similarity. Whether APL's **activity** reshapes the KC representation is the question of the model, from Report 05 on.

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6 Odor-by-odor distance between response-weighted synapse maps

Yang asked for the "correlation (euclidean) between odors by weighing the APL ↔ KC/PN synaptic coordinates by responses". The reading used here: each odor turns APL's synapses into a map over the calyx, every synapse e at position $\mathbf{x}_e$ carrying weight $w_o(e)$, smoothed by a 3D Gaussian of width σ. Two odors' maps are then compared by their overlap:

$$K_{ab} \;=\; \frac{1}{N}\sum_{e}\sum_{e'} w_a(e)\, G_{ee'}\, w_b(e'), \qquad G_{ee'} = \exp\!\Big(-\frac{|\mathbf{x}_e - \mathbf{x}_{e'}|^2}{4\sigma^2}\Big) \tag{5}$$

N = number of synapses with a response. $G_{ee'}$ is the overlap of two Gaussians of width σ, scaled to 1 at distance 0. σ = 0 means G = identity: each synapse compared only with itself.

$$d_{ab} \;=\; \sqrt{K_{aa} + K_{bb} - 2K_{ab}} \tag{6}$$

Euclidean distance between the two smoothed maps.

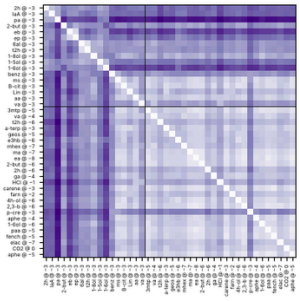
$$S_{ab} \;=\; \frac{K_{ab}}{\sqrt{K_{aa}\,K_{bb}}} \tag{7}$$

Cosine similarity of the two smoothed maps.

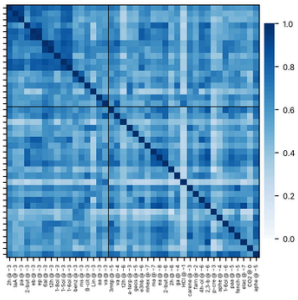
The reference is "responses only": the same quantities with glomeruli counted once and no positions. For KC synapses (up to about 21,000 per calyx) the matrix G is too large to hold at once, so K is summed in blocks of rows.

ORN, megamat + diagnostics (dataset 1, 9 flies): odor-by-odor distance (purple) and cosine similarity (blue) between response-weighted maps of PN ↔ APL calyx synapses
Each synapse carries its glomerulus's response to the odor; maps smoothed with a 3D Gaussian of σ = 2 µm (corrected calyx). Lines separate the megamat odors (top left) from the diagnostics. Each panel has its own color scale.

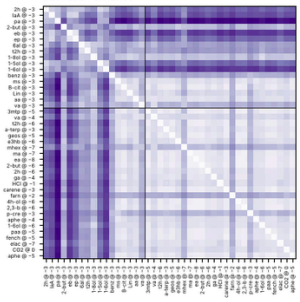
responses only (glomeruli once, no positions)
Euclidean distance



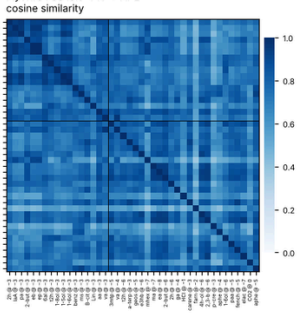
responses only (glomeruli once, no positions)
cosine similarity



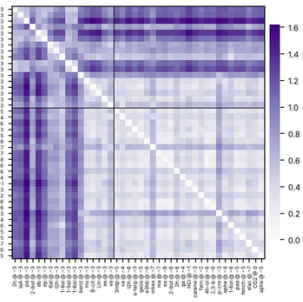
FlyWire 783 left · PN → APL
Euclidean distance



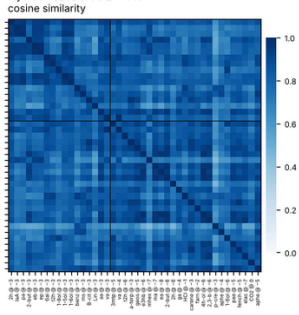
FlyWire 783 left · PN → APL
cosine similarity



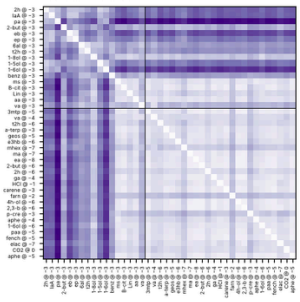
FlyWire 783 left · APL → PN
Euclidean distance



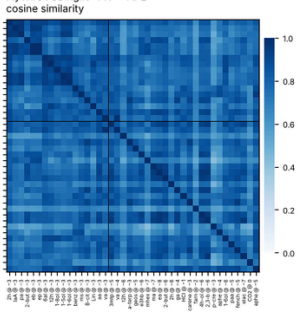
FlyWire 783 left · APL → PN
cosine similarity



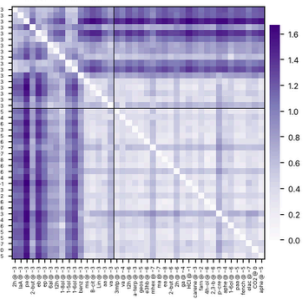
FlyWire 783 right · PN → APL
Euclidean distance



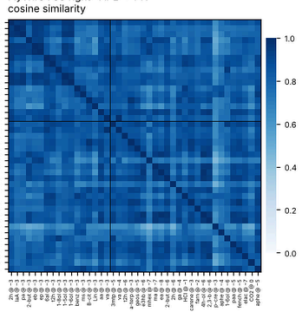
FlyWire 783 right · PN → APL
cosine similarity



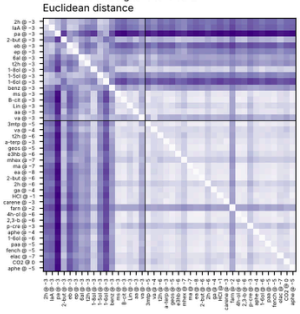
FlyWire 783 right · APL → PN
Euclidean distance



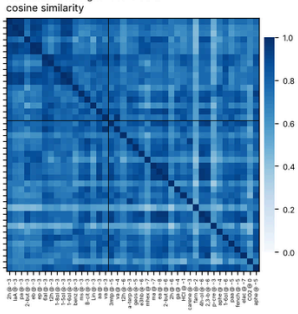
FlyWire 783 right · APL → PN
cosine similarity



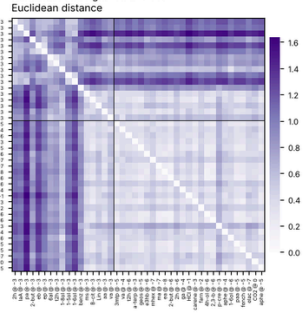
Hemibrain v1.2 right · PN → APL
Euclidean distance



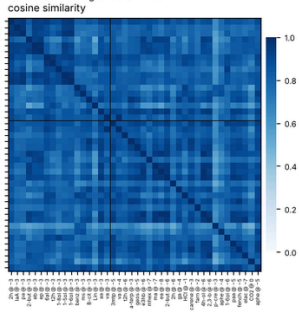
Hemibrain v1.2 right · PN → APL
cosine similarity



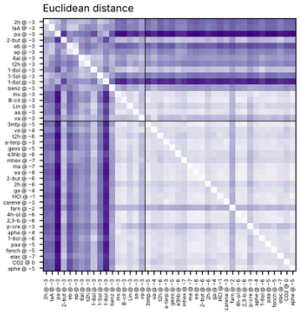
Hemibrain v1.2 right · APL → PN
Euclidean distance



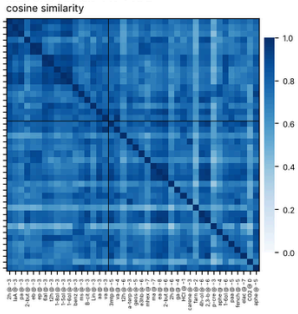
Hemibrain v1.2 right · APL → PN
cosine similarity



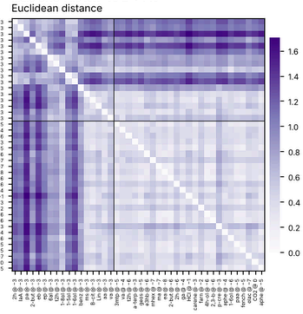
Male CNS v1.0 left · PN → APL
Euclidean distance



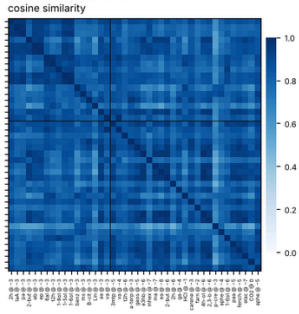
Male CNS v1.0 left · PN → APL
cosine similarity



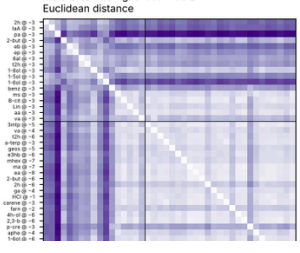
Male CNS v1.0 left · APL → PN
Euclidean distance



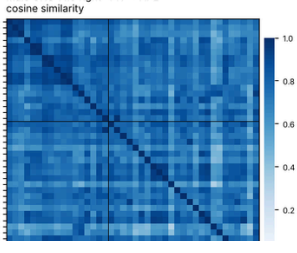
Male CNS v1.0 left · APL → PN
cosine similarity



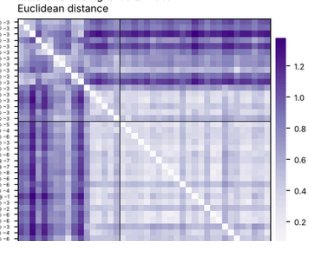
Male CNS v1.0 right · PN → APL
Euclidean distance



Male CNS v1.0 right · PN → APL
cosine similarity



Male CNS v1.0 right · APL → PN
Euclidean distance



Male CNS v1.0 right · APL → PN
cosine similarity

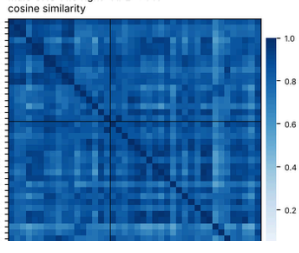




Figure 8. Odor-by-odor distance and similarity of PN ↔ APL synapse maps. ORN dataset 1 (megamat + diagnostics). Top row: responses only. Then one row per calyx (FlyWire L and R, hemibrain R, male CNS L and R). Columns: PN → APL and APL → PN, each as Euclidean distance d_{ab} (purple) and cosine S_{ab} (blue), with σ = 2 µm. Lines separate the megamat odors (top left) from the diagnostics. Each panel has its own color scale. Dataset 4 is in `with_outside_ROI_synapses/odor_weighted/new_ORN_2026-11-03/ORN_valid2_diag_odor_distance_heatmaps_PN_APL.png`.

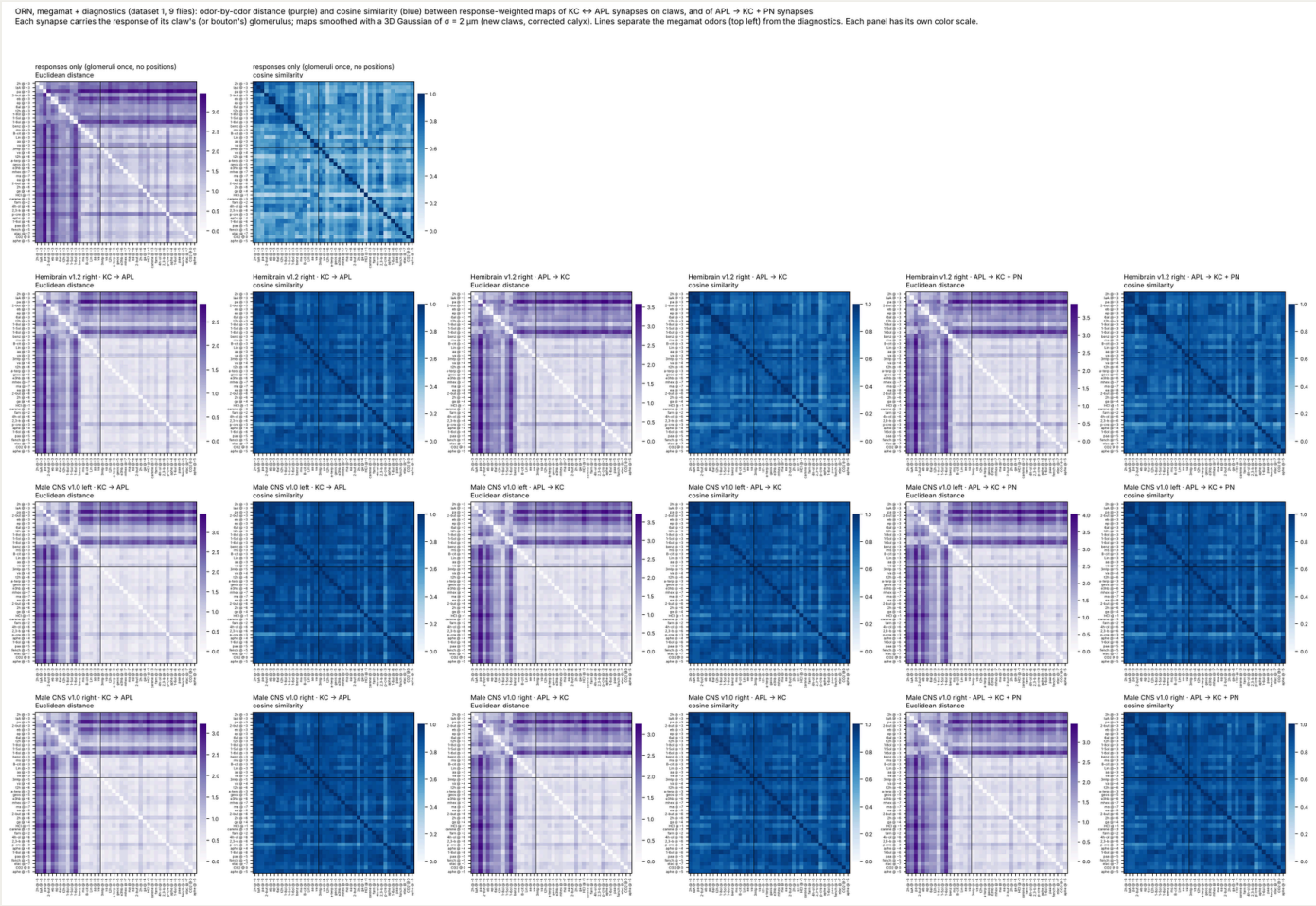


Figure 9. Odor-by-odor distance and similarity of KC ↔ APL and APL → KC + PN synapse maps. As Figure 8 for the three model calyces, with columns KC → APL on claws, APL → KC on claws and APL → KC + PN. Dataset 4 is in `ORN_valid2_diag_odor_distance_heatmaps_KC_APL.png`.

Table 6. How closely each calyx's odor matrix follows the responses-only one. Correlation over odor pairs between a calyx's matrix and the responses-only matrix; range over calyces and directions.

ORN	Synapses	distance, σ 0	σ 2 µm	σ 5 µm	cosine, σ 0	σ 2 µm	σ 5 µm
dataset 1	PN ↔ APL (5 calyces)	0.95–0.98	0.93–0.96	0.90–0.92	0.79–0.90	0.70–0.77	0.51–0.60
dataset 1	KC ↔ APL on claws (3 calyces)	0.98–0.99	0.94–0.96	0.90–0.92	0.92–0.95	0.76–0.83	0.60–0.63
dataset 1	APL → KC + PN (3 calyces)	0.98–0.98	0.94–0.95	0.90–0.91	0.91–0.93	0.76–0.82	0.59–0.59
dataset 4	PN ↔ APL (5 calyces)	0.94–0.99	0.91–0.97	0.86–0.92	0.87–0.97	0.80–0.90	0.67–0.79
dataset 4	KC ↔ APL on claws (3 calyces)	0.96–0.98	0.92–0.95	0.87–0.90	0.92–0.96	0.82–0.89	0.70–0.75

ORN	Synapses	distance, σ 0	σ 2 µm	σ 5 µm	cosine, σ 0	σ 2 µm	σ 5 µm
dataset 4	APL → KC + PN (3 calyces)	0.96–0.97	0.92–0.95	0.87–0.90	0.92–0.95	0.82–0.89	0.69–0.73

- **Euclidean distances mostly follow response strength.** An odor that drives many glomeruli strongly is far from every other odor. At σ = 2 µm the calyx matrices match the responses-only one at r 0.93–0.96 (dataset 1).
- **Cosine similarities depend more on where the synapses sit.** Without smoothing they follow the responses-only matrix. With smoothing they drift away from it (r 0.51–0.63 at 5 µm, dataset 1), because the Gaussian mixes neighboring boutons and claws of different glomeruli.
- KC synapse maps follow the responses-only matrix more closely than PN maps, already without smoothing (σ = 0), where only how the synapses are spread over glomeruli matters.

Open interpretation

"Correlation (euclidean)" was read as Euclidean distance and cosine between smoothed maps (both are shown). If Yang meant something else, for example a correlation over a voxel grid, the same inputs can be reused.

Caveats

Synapse counts stand in for strength; nothing here measures how strong a single synapse is.

The ORN units of `responses_wide.csv` are not stated in the file. They are used as given.

FlyWire and BANC KCs have no new claws (the new rule needs KC skeletons, done for hemibrain and male CNS), so the KC maps cover the three calyces of the model.